

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data (Hannah Ritchie, Max Roser, Pablo Rosado)

75% of Global GHG Emissions Still Come from Fossil Fuels

The Urgency

- Since the Industrial Revolution, energy systems have been dominated by fossil fuels.
- 75% of global greenhouse gas emissions come from burning fossil fuels for energy.
- At least 5 million premature deaths per year are caused by local air pollution from fossil fuel combustion.

The Solution

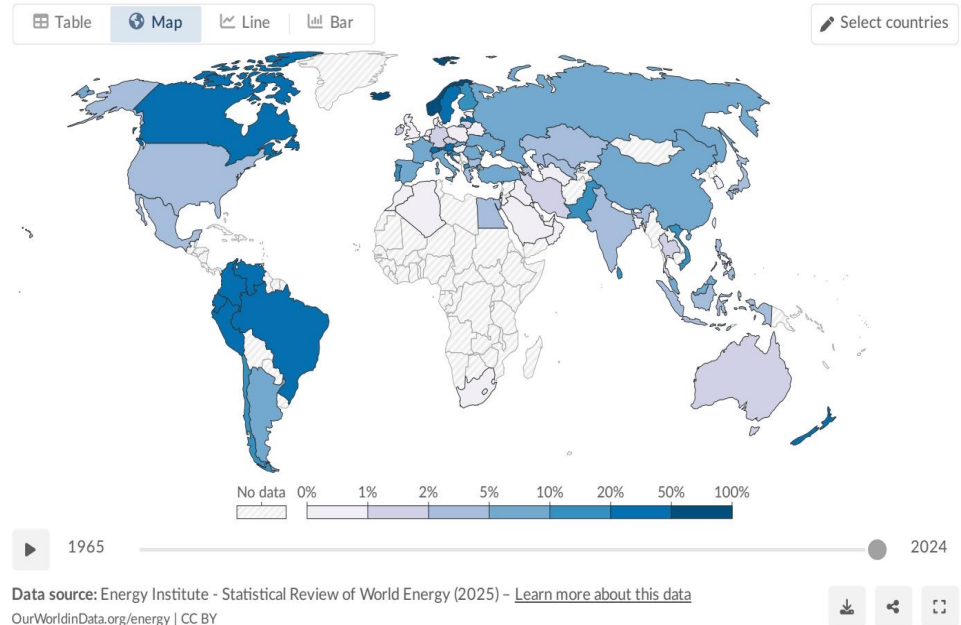
- The world must rapidly shift toward low-carbon energy sources.
- Renewable energy technologies are scaling quickly, including:
 1. Solar
 2. Wind
 3. Hydropower

Renewables Now Supply ~14% of Global Primary Energy

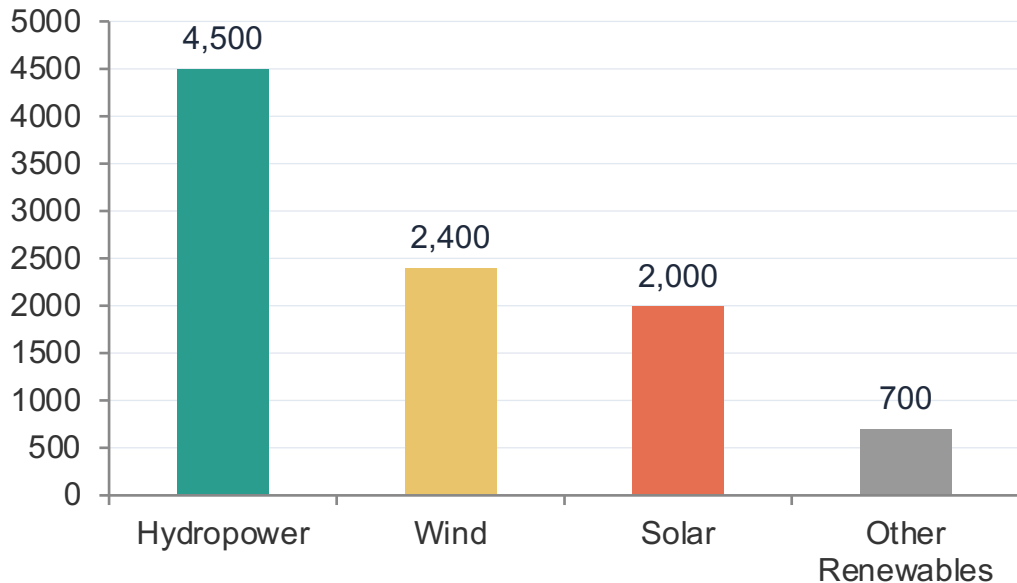
- Approximately one-seventh (~14%) of the world's primary energy now comes from renewable technologies (2024, substitution method).

Current Milestone

- Primary energy includes electricity, transport, and heating.
- Transport and heating remain harder to decarbonize.
- Regional variance: Norway, Brazil, and New Zealand have high shares; Middle Eastern and some Asian countries remain below 5%.



Renewable Generation Is Growing Rapidly, Led by Wind and Solar



Acceleration:

Wind and solar generation have accelerated dramatically since ~2010.

Matching Hydro:

Combined, wind and solar now nearly match hydropower's massive 4,500 TWh output.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

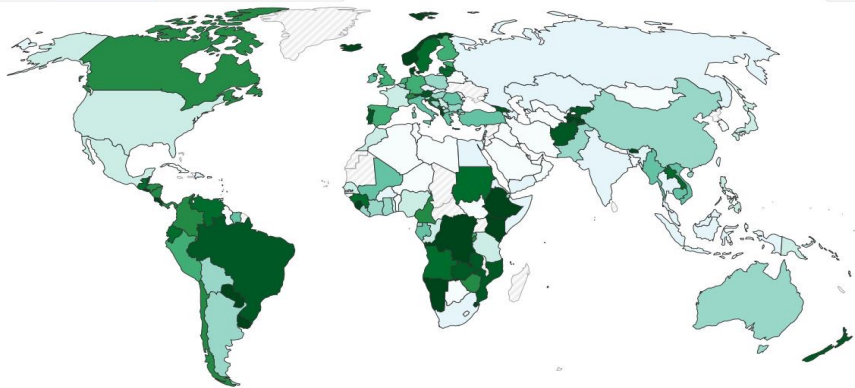
Table

Map

Line

Bar

Select countries



No data 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

1985

2025

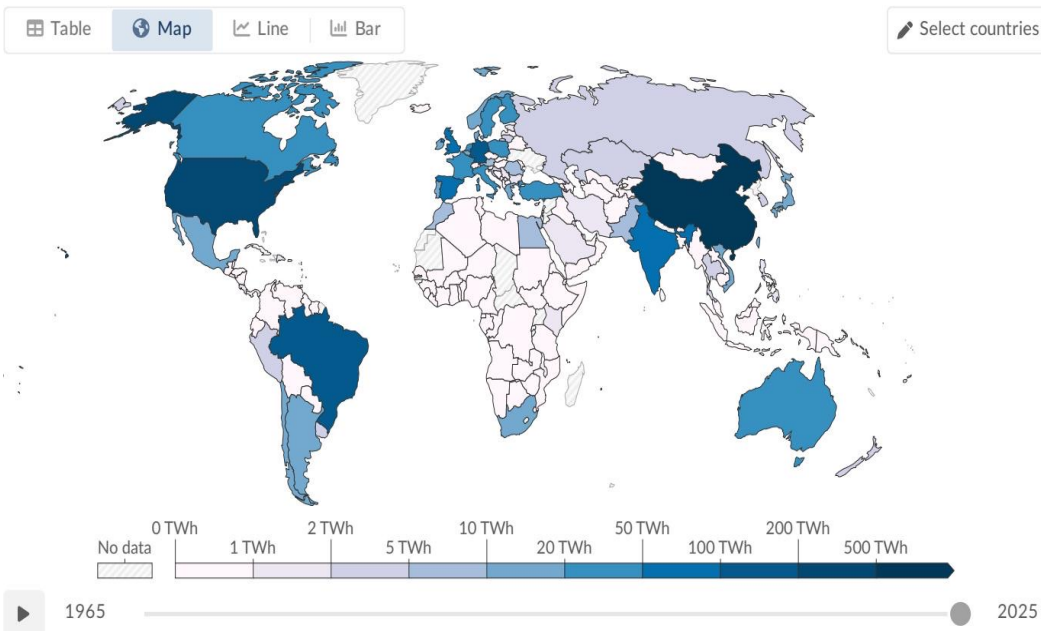
Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Renewable Share by Region

Country	Share (%)
Norway	98%
Brazil	85%
Canada	65%
China	35%
India	25%
Saudi Arabia	9%

Substantial progress in electricity (coal and oil primary energy 9%), but huge gaps persist between leaders and laggards.

Hydropower Remains the Largest Renewable Source at ~50% of Generation



Dominant Source

Accounts for roughly half of all renewable electricity globally.

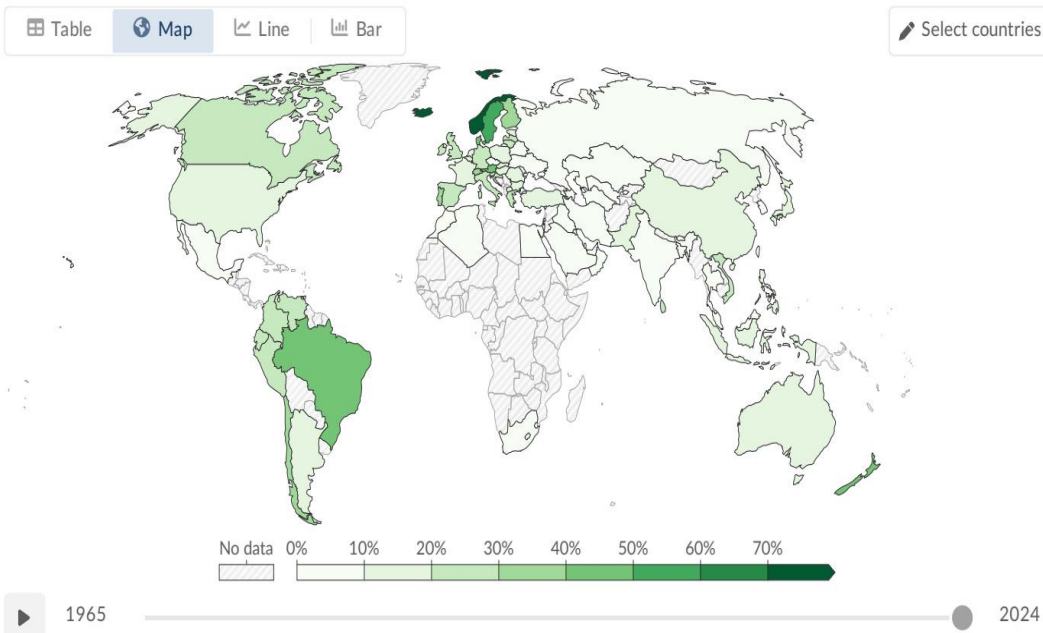
Long-standing Role

Has generated electricity at scale for over a century.

Top Producers

China, Brazil, Canada.

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind vs. Solar (2024)

- Wind: ~2,400 TWh (~25% of renewable mix)
- Solar: ~2,000 TWh (~21% of renewable mix)

Rapid Acceleration

Solar is the fastest-growing source, driven by falling costs and policy support since 2010.

Catching Up

Combined, they now nearly match hydropower output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia & LatAm Lead

Scandinavia & LatAm

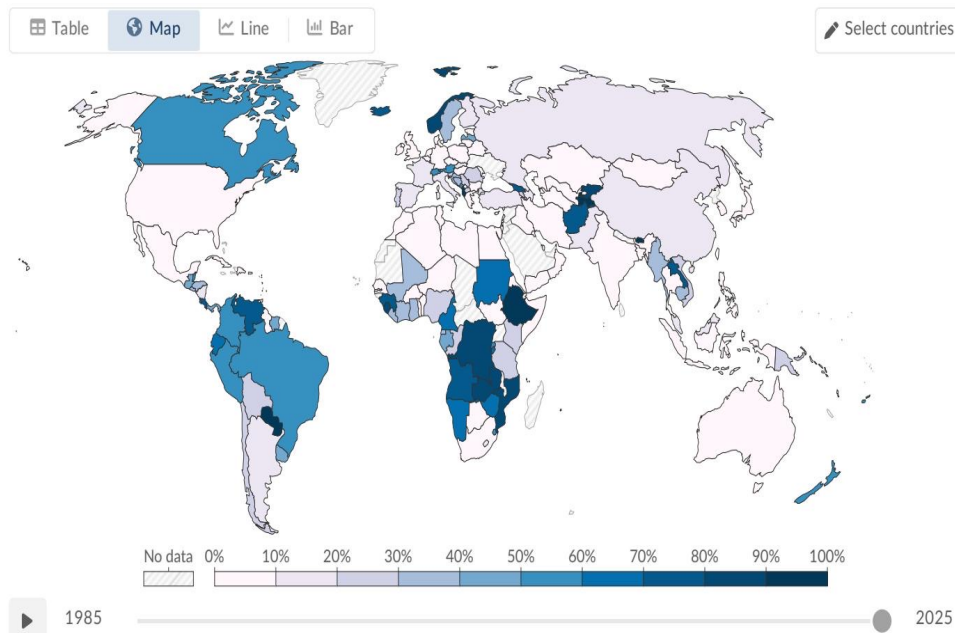
Scandinavia, Brazil, and Canada lead at 65–98% renewable electricity, benefiting from abundant hydropower.

Europe

Nations like Denmark and Germany have invested heavily in wind and solar.

Africa & Middle East

Many countries in these regions remain below 10% renewable electricity. Middle East relies heavily on fossil fuels, while Sub-Saharan Africa has minimal renewable infrastructure.



Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

Milestones Reached: Renewables now supply ~14% of global primary energy and ~30% of global electricity.

Technology Leaders: Hydropower is still dominant, but wind and solar are the fastest-growing and rapidly catching up.

Uneven Transition: Stark regional disparities remain; while some countries exceed 90%, others are below 10%.

Sector Gaps: Transport and heating lag far behind electricity in decarbonization.

The Path Forward:

Although renewable growth is historic, the transition must accelerate significantly to meet climate targets.